

JINGHUA YAO Ph.D.

Principal Applied Scientist | Generative AI & Machine Learning

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Professional Summary

Applied scientist with experience at Microsoft, Apple, and Samsung, combining a Ph.D. in mathematics with hands-on development of enterprise AI systems. Expertise spans LLM agents, distributed fine-tuning, multimodal learning, NLP, time-series forecasting, and computer vision. Track record of architecting applications, automating operational workflows, and translating mathematical modeling into practical machine learning solutions.

Technical Expertise

Generative AI LLM agents, LLM/VLM fine-tuning, transformers, prompt engineering, document Q&A, summarization, multimodal understanding, embeddings

Machine Learning NLP, sentiment analysis, forecasting, anomaly detection, CNNs, ensemble methods, statistical modeling, optimization

Engineering Python, PySpark, C#, TypeScript, Go, scikit-learn, Docker, AWS, Amazon S3, Splunk, Git, Streamlit, Gradio

Professional Experience

Microsoft

California

Principal Applied Scientist

Jul 2025 – Present

- Develop LLM agent workflows involving memory, reasoning, actions, and reflection for AI applications.
- Develop presentation-generation workflows and evaluate outputs for AI applications.

Apple

California

Senior Machine Learning Engineer

May 2020 – Jun 2025

- Architected and implemented visual understanding with late interaction for enterprise applications.
- Developed distributed fine-tuning workflows for LLMs and vision-language models, including Mistral, Florence-2, Phi-3.5, and Molmo, supporting embedding, Apple Support, captioning, and generative AI applications.
- Built enterprise LLM applications for customer sentiment analysis, contextual Q&A, document analysis, and summarization, improving feedback resolution and reducing labor and device costs.
- Architected and implemented automated survey-health reporting using Splunk, Amazon S3, anomaly detection, and email notifications, replacing manual team workflows.
- Built time-series forecasting, trend analysis, and anomaly detection systems for call centers and survey services; supported improvements in call-answer rates.
- Developed transformer-based sentiment analysis and classification for customer comments.
- Delivered ML and AI technical lectures as a main contributor to a series of 32+ one-hour sessions on modeling and algorithms.

Professional Experience (continued)

Samsung Display American Lab

San Jose, CA

Staff Machine Learning Engineer

Mar 2019 – May 2020

- Led problem formulation and development of a display-defect detection and classification system used in products.
- Combined convolutional neural networks, dimensionality reduction, and classical ML using approximately 800 MAP and TEM images; developed a product-used classification solution.

Selected Additional Projects

- **Predictive modeling for lending (2018):** Built a random-forest model for a small, imbalanced loan-application dataset; compared its performance with SVM and logistic-regression baselines.
- **Salary prediction (2017):** Built an ensemble application combining linear regression and random forests using approximately **one million** job records.

Research & Academic Experience

Kent State University

Kent, OH

Research Scholar, Department of Mathematics

Sep 2016 – Sep 2017

The University of Iowa

Iowa City, IA

Visiting Assistant Professor, Department of Mathematics

Aug 2013 – Jan 2016

Research in nonlinear analysis, partial differential equations, dynamical systems, and variational methods. Author or coauthor of **18 journal publications**, including work in *Physica D*, *Journal of Differential Equations*, and *Communications in Mathematical Sciences*.

Selected Publications

- Yao, J., Liu, D., and Wang, X. “Patterns of complex oscillations and instability in chemical reactions.” *Communications in Mathematical Sciences*, 2019.
- Dai, G., Yao, J., and Li, F. “Spectrum and bifurcation for semilinear elliptic problems in $\mathbb{R}^N$.” *Journal of Differential Equations*, 2017.
- Yao, J. “ $O(2)$ -Hopf bifurcation for a model of cellular shock instability.” *Physica D: Nonlinear Phenomena*, 2014.

Education

Ph.D. in Mathematics, Indiana University Bloomington, USA

2013

Master's in Mathematics, Lanzhou University, China

2008

Bachelor's in Mathematics, Lanzhou University, China

2005

Professional Service & Recognition

Invited reviewer for *Mathematical Reviews*, American Mathematical Society; journal referee for *Scientific Reports*, *Journal of Mathematical Analysis and Applications*, and other mathematical journals.

National Prize and Provincial Special Prize, National Mathematical Modeling Contest, 2004.